Next phase plan: tasks, 10-seed costs and recalibration

Feasible Rank Window project · prepared 25 October 2026 · based on the frw-v4 results (1,088 runs, docs/FINDINGS_V4.md)

Why change the tasks

On RTE, MNLI and SST-2, LoRA rank 1 already learned every task to within about 1 point of the best rank (all four models), so accuracy was flat across ranks. These classification tasks have a very low intrinsic dimension (Aghajanyan et al., ACL 2021). A rank window can only show real edges on tasks that need more adaptation capacity, so the next phase mixes low- and high-capacity NLP tasks in one sequence.

1. Proposed task sequence

#	Task	Dataset	Capacity needed	Train / eval	Metric
1	Topic classification	AG News (4 classes)	low	4,000 / 500	accuracy
2	Natural language inference	MNLI (link to current results)	low to medium	4,000 / 500	accuracy
3	Extractive question answering	SQuAD v1.1	medium	4,000 / 500	exact match / F1
4	Dialogue summarisation	SAMSum	high	4,000 / 500	ROUGE-L
5	Maths word problems (step by step)	GSM8K	high	4,000 / 500	exact match (final answer)

• **One format for every task:** the model generates the answer as text through its own output layer. This removes the per-task classifier heads, which caused most of the measured forgetting in v4.

• **Equal data:** the same 4,000 training examples per task; a fixed 500-example evaluation subset per task keeps the cost of text generation under control.

• **General-knowledge check:** a fixed 500-question MMLU subset (answer-likelihood scoring) after each task, to measure forgetting of pretrained knowledge as well as of earlier tasks.

• **Two task orders:** easy to hard, and the reverse.

• **Hypothesis to test:** the window stays at r ≈ 4–32 for AG News and MNLI but moves upward for SAMSum and GSM8K, i.e. the feasible window depends on the type of task.

2. Cost estimate with 10 seeds (Modal L40S)

Scaled from measured v4 run times: about 7× the training tokens per run (5 tasks × 4,000 examples at ~150 tokens versus 3 × 2,490 at ~55), about 5 minutes of text-generation evaluation per run, and about 30% extra for LoRA on all linear layers. L40S at $2.17 per hour including CPU and memory.

Model	Per run	10 seeds × 6 ranks (1, 4, 16, 64, 256, 1024)	LR search (3 ranks × 4 LRs × 2 seeds)	Total
Llama-3.2-1B	~$0.80	~$48	~$15	~$63
Qwen2.5-1.5B	~$1.20	~$72	~$23	~$95
Granite-3.3-2B	~$2.00	~$120	~$38	~$158
Qwen2.5-3B	~$2.00	~$120	~$38	~$158
All four models				**~$475**

Schedule against $180 per month	Models	Estimated cost
Month 1	Llama-3.2-1B + Qwen2.5-1.5B	~$158

Schedule against $180 per month	Models	Estimated cost
Month 2	Granite-3.3-2B	~$158
Month 3	Qwen2.5-3B	~$158
Optional	Second task order	+60–70%
One-time extras	Epoch check (~$15); frozen-model, head-only baseline (under $5); pilot (~$5)	~$25

Uncertainty: treat every figure as ±40%. Run a ~$5 pilot first to measure the real time per run, exactly as the GPU benchmark was done before v4. The six accounts keep the same safeguards: admission cap below $30, a hard stop on the real bill at $29, and an end-of-month deadline.

3. Will the hyperparameters need recalibrating?

Yes, but only a few, and cheaply. The costs above already include the learning-rate search.

Hyperparameter	Recalibrate?	Reason
Learning rate	Yes (required)	The strongest driver of forgetting in every experiment. The loss changes (generating text instead of classifying), the tasks change, and the all-linear selection already differed from q/v. Search at 3 ranks (4, 64, 1024), 4 LRs, 2 seeds, and reuse it for the other ranks: v4 showed the selected LR barely depends on rank when α is fixed.
Epochs per task	Yes (quick check)	3 epochs was confirmed for classification only (E3). Generative tasks may need a different number: one check at r = 16 comparing 2 / 3 / 4 epochs with 5 seeds, about $15.
Max sequence length	Yes	Raise from 256 to 512 tokens: SQuAD, SAMSum and GSM8K inputs plus answers exceed 256.
Evaluation decoding	Set, not tuned	Greedy decoding with a fixed maximum length per task (e.g. 8 tokens for classification, 64 for summaries, 256 for maths).
Classifier head learning rate	No longer needed	The generative format has no classifier heads.
α = 32 (scale α/r), AdamW (0.9, 0.999, 1e-8), batch 16, warmup 6% + cosine, clipping 1.0, weight decay 0, dropout 0	No	Conventions or settings validated in v4. Re-tuning them adds cost without clear benefit.
Target modules	No (decided)	LoRA on all linear layers, as agreed after the E2 experiment.
Seeds	No (decided)	10 seeds per reported setting; 2 tuning seeds kept separate from reported seeds.

Sources

Measured costs and timings: results-v4/analysis (runs.csv, compute_by_model_and_stage.json). Findings: docs/FINDINGS_V4.md, sections 1–6. Intrinsic dimension: Aghajanyan, Zettlemoyer & Gupta (2021), "Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning", ACL. Pricing: Modal list prices on 25 October 2026 (L40S $1.95/h plus CPU and memory).